Building Blocks of Tabletop Game Design

An Encyclopedia of Mechanisms

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CRC Press
Taylor & Francis Group
Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business

CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

Printed on acid-free paper

International Standard Book Number-13: 978-1-138-36549-0 (Paperback)
978-1-138-36552-0 (Hardback)

Library of Congress Cataloging-in-Publication Data

Names: Engelstein, Geoffrey, author. | Shalev, Isaac, author.
Title: Building blocks of tabletop game design : an encyclopedia of mechanisms / Geoffrey Engelstein and Isaac Shalev.
Description: Boca Raton, FL : Taylor & Francis, 2019. | Includes bibliographical references.
Identifiers: LCCN 2019010857| ISBN 9781138365490 (pbk. : alk. paper) | ISBN 9781138365520 (hardback : alk. paper)
Subjects: LCSH: Board games--Design and construction.
Classification: LCC GV1312 .E64 2019 | DDC 794--dc23
LC record available at https://lccn.loc.gov/2019010857

Visit the Taylor & Francis Web site at
http://www.taylorandfrancis.com

and the CRC Press Web site at
http://www.crcpress.com

Geoff:

To Susan, for a wonderful 40 years, and to our children, Brian and Sydney, for always challenging my strategies across the game table, and challenging my ideas across the kitchen table.

Isaac:

To my wife, a quick nod of appreciation.
Let's not make a scene about this.

To my children, Jesse, Alex, and Maya, you are
the gaming partners of a lifetime.

To my design partner, Matt Loomis, from whom I learned
more than I could contain in this slim volume.

Lastly, to my brother, Kuty Shalev, my constant gaming partner,
my first DM, my first co-designer, my Kickstarter patron and
my coach, mentor, and cheerleader in all my pursuits.

Contents

Foreword

A game is a language.

In a way, when you sit down with someone to play *Chess*, *Poker*, or *Cosmic Encounter*, you are beginning a conversation.

Each move you make in the game is a way for you to express yourself to your partner—a way for you to make meaning. Every reckless aggression and coy bluff, every greedy power grab and defensive stall for time is a word, a phrase, a statement in the language of the game.

These conversations we have with each other, as we play games, can be awkward and confusing. Or deliciously cruel. Or wildly creative and unexpectedly beautiful. The kind of conversation that tickles parts of your mind that you didn't know were there, the kind of conversation that sticks with you, turns acquaintances into bitter rivals, and later into the best of friends.

But a game is not just a language.

It's more complicated than that. The moves we make as we play are a language built on top of yet another language. Beneath the meaning we express through the moves of the game is a kind of hidden grammar.

You move a pawn, pass a ball to a teammate, or maneuver your virtual avatar: all of them are statements made through gameplay. But what are the structures that make *those* meanings possible? The rules that permit action? The blood and bones and spirit that animates the body of the game as it is played?

That, exactly, is the subject of *Building Blocks of Tabletop Game Design*. So prepare yourself. Between these covers lies the hidden grammar of games. The structures behind the structures. The DNA of fun.

This book is a dictionary for the secret language of games.

You see, although games and play are ancient human endeavors (the Egyptian game *Senet* is at least 5,500 years old), it's only in the twentieth century that games shifted from folk culture to authored media. And as tabletop games, role-playing games, and video games have become industries of their own, we've done our best to understand them.

If you are reading this book, you are likely to be a serious game player. Like many others, you play games, discuss them, dissect them, and sometimes even design new ones. And for many years, many smart designers, critics, players, and scholars have been searching for the right way to talk about games, to understand how they work, and to figure out how best to approach the creation of new designs.

That's why *Building Blocks of Tabletop Game Design* is so significant. The incredibly detailed pages that follow, pages that crack open the inner components of tabletop games, constitute a kind of *Rosetta Stone* for game grammar.

Make no mistake. This book is a big deal.

Building on their own impressive accomplishments as designers, their relentless intellectual curiosity, and a seemingly limitless connoisseurship of analog and digital games, Geoff and Isaac have put together a tremendously rigorous, wonderfully insightful, and astoundingly accessible encyclopedia of the elements that make tabletop games tick.

Why is this project so important? As someone who has studied, designed, and taught game design for 25 years, this volume is the sort of book I realize now I always needed. A manual for game mechanics. A foundation for structural analysis. An inspiration for new ideas. A sourcebook for teaching design.

On its surface, *Building Blocks of Tabletop Game Design* might seem like a deeply geeky endeavor, an obsession with rules and structures. The kind of overly formalist ludology that has (rightfully) gotten into so much trouble in recent years. And in part it is. But look deeper. In games, what seems like pure abstract structure is inseparable from human experience. Chapter 4, on resolving game actions (those things players do to make meaning) begins in statistics and the mathematics of randomness. But then it moves into the thorny terrain of the prisoner's dilemma, engaging with psychology and diplomacy and even ethics—structures that help shape how people treat one another.

Reading any one of the dozens of modules in this book is like lifting up a big mossy rock in the forest. Underneath, there's an unexpected universe of complexity, a miniature ecosystem of moving parts, just waiting to be discovered.

The language of games is everywhere.

All of this gushing praise is well deserved. This book is a landmark in the study of games. But don't let that fool you. As enchanting as it is, *Building Blocks of Tabletop Game Design* is not a magic bullet that is going to suddenly let you understand exactly how every game works, become a tournament winner, or guarantee that your next design will be a hit.

The hidden grammars that this book describes, the structures behind the structures, are only part of the picture. *Building Blocks of Tabletop Game Design* doesn't investigate how games tell stories, or impact the lives of their players, or embody ideological values, or fit into larger cultural landscapes, or even how they might make a profit. That's just not what this book is about. And that's perfectly all right. Its titanic strength comes from its incredibly tight focus on the fundamental elements of games.

Despite these limitations, is this book still useful? Hell yes. I was recently trying to design a bidding structure for a game, and found myself jaw-droppingly enlightened, reading Chapter 8 on Auctions, which details no less than 16 distinct bidding structures. Without a doubt, I will use this text in my classes. I am planning, for example, to assign a project by having each student turn to two random pages in this book and then create a game that combines both mechanics.

But there's more. The insights and ideas in *Building Blocks of Tabletop Game Design* can be applied outside of games.

How can we resolve an argument between two entrenched opposing positions? How should a winner in a multicandidate election be determined? What economic incentives lead to the best distribution of wealth? What's the right way to ensure fairness for all? These are dilemmas of modern society. And they are also the kinds of problems with which game designers wrestle on a daily basis. Believe it or not, the answers to these deeply important questions might begin to be answered by looking at the structures within structures of this book.

Lastly, games are beautiful.

Don't get me wrong. I don't want to leave you with the impression that this book is valid only because it is useful. It doesn't matter that *Building Blocks of Tabletop Games* might be used to help teach classes, or even make society better. This book is important because it is a heartfelt and soul-enriching love letter to games.

It is a lens to help us see the games we adore with fresh eyes. An advanced seminar in the complexity of systems. A spell book filled with recipes for creative play. *Building Blocks of Tabletop Game Design* unpacks the mystery of how these nerdy, knotty, collections of rules—boxed up with cards, dice, and colorful tokens—produce something which is in fact the very opposite of rules.

For those ready to appreciate the beauty of games, the joy leaps off of every page. Isaac and Geoff have given us whole new ways to have conversations with the people and the games we love.

Happy reading.

Eric Zimmerman
Game Designer and Arts Professor,
NYU Game Center,
New York City, April 2019

Acknowledgments

Thank you to Erik Zimmerman and Doug Maynard for your excellent feedback and insightful comments on the manuscript. The readers thank you even more! Rob Daviau, Tim Fowers, Mark Herman, and Sen-Foong Lim provided key early enthusiastic support, without which we would not have embarked on this project.

This work would also not exist if not for those who blazed this trail and whose writings have shaped our thinking. Greg Costikyan's *I Have No Words & I Must Design: Toward a Critical Vocabulary for Games* (Tampere University Press, 2002) was Isaac's introduction to the world of critical thought about games. *Design Patterns: Elements of Reusable Object-Oriented Software* (Addison-Wesley, 1994) by Erich Gamma, John Vlissides, Ralph Johnson, and Richard Helm showed Geoff how design concepts could be broken down and built back up again. *Rules of Play* (MIT Press, 2003) by Katie Salen and Erik Zimmerman was another foundational text. We would also like to recognize the books, lectures and blog posts by Jesse Schell, Bruno Faidutti and Mark Rosewater, who continue to contribute to the growing body of thinking about games and game design.

Daniel Solis created the delightful diagrams that illustrate each mechanism. His ability to concisely capture complex concepts is uncanny, and this book benefits greatly from his participation.

We are also indebted to Donald Dennis, Erik Dewey, Seth Jaffee, TC Petty III, Dirk Knemeyer, Adrienne Ezell, Sen-Foong Lim, Ben Begeal, Raph Koster, James Ernst, Eric Lang, Ralph Anderson, Stephanie Straw, Nicole Kline, Anthony Amato, Jr., Gil Hova, Ryan Sturm, Mike Fitzgerald, Tom Vasel, Eric Summerer, all the fans and listeners of the *On Board Games* and *Ludology* podcasts, and our boardgame Twitter family. This work is, if nothing else, a memento of our relationships, conversations, and friendships. Thank you.

Finally, Sean and Jessica at Taylor & Francis Group were absolutely stellar to work with, and deftly handled two demanding authors.

Authors

Geoffrey Engelstein is the designer of many tabletop games, including *The Ares Project*, the *Space Cadets series*, *The Dragon & Flagon*, *The Expanse*, and more. He is the host of Ludology, a bi-weekly podcast about game design in its seventh year, and a ten-year contributor to the *Dice Tower* podcast with his bi-weekly "GameTek" segments discussing the math, science, and psychology of games. He has published *GameTek: The Math and Science of Gaming* (2018), which was re-published as *GameTek: What Games Can Teach Us about Life, the Universe and Ourselves* by Harper-collins in early 2019. He is on the faculty of the NYU Game Center as an adjunct professor for board game design, and has been invited to speak about game design at PAX, Gen Con, Metatopia, and the Game Developers Conference. When not talking about, designing, or playing games, Geoff runs Mars International, a product development firm focusing on consumer and medical device engineering. He has a BS in physics and a BS in electrical engineering from the Massachusetts Institute of Technology.

Isaac Shalev is the designer, together with his longtime collaborator Matt Loomis, of a variety of board games including *Seikatsu, Ravenous River*, and *Show & Tile*. He is also the co-host of *On Board Games*, one of the longest running and most-respected tabletop game podcasts. Over the last five years, Isaac has interviewed people from every corner of the industry, including top designers, publishers, distributors, illustrators, convention runners, store owners, and more, and brought their insights to the gaming community at large. Isaac also writes about game design at www.kindfortress.com, and his series on tabletop game design patterns is a favorite among game designers. Isaac's eclectic work in the game industry includes advising publishers, editing rulebooks, speaking about game design at schools, running board gaming fundraisers, and organizing PACT—the

Playtesters Alliance of Connecticut. When he's not playing games, making games, or talking about games, Isaac runs Sage70, Inc., a data strategy consultancy that works exclusively with nonprofit organizations. Isaac lives in Stamford, Connecticut, with his wife, three kids, a dog, and a lawn that grows unreasonably quickly.

Introduction

There is no better time to be a tabletop board gamer. Hobbyists will debate whether the mid-1990s, which saw the rise of *The Settlers of Catan* and *Magic: The Gathering*, was the golden age of gaming, or whether today is that golden age. No matter what your personal view may be, there is no doubt that modern tabletop gaming is in a period of exceptional fertility and is flourishing.

Games have come a long way since humans invented dice, pawns, and boards, and archeologists regularly turn up these artifacts at sites dating back 6000+ years ago. The standard deck of cards has its beginnings over 1000 years ago, and with the advent of the printing press, games became an ever-cheaper and more ubiquitous luxury. The history of the last century or so of games encompasses mass-market titles including *Monopoly*, *Scrabble*, *Clue*, and *Trivial Pursuit*, as well as the emergence of a whole new genre of tabletop games. The genre is so new that a single name has yet to meaningfully describe it, though suggestions have included hobby games, designer games, and other, stranger terms like TGOO ("these games of ours"). No matter what you call them, today's games feature higher levels of player agency, meaningful decisions, and substantially improved production values, and offer something entirely new to players, as compared to mass-market games.

Others have written about how war games developed into modern games in America, and the countervailing rise of a European design sensibility in the post-World War II period that rejected war as a setting, and direct conflict as a game dynamic. Our goal here is different. We'd like to address not how games evolved, but how games are designed.

Even as gaming itself has taken flight, and even as thousands of new games are published each year to an ever-expanding audience of gamers, one critical aspect of gaming remains in a nascent stage: the art and craft of game design itself. Games have taken great advantage of modern communication networks and methods to spread, but we encounter challenges in spreading the knowledge and skills needed to design these games. In substantial part this is because we do not yet have a strong shared vocabulary for discussing design.

In creative industries, from literature to film to video games, practitioners, participants, and critics develop a shared language, a common reference library, and a set of skills, approaches, and techniques for their field. Film directors learn the difference between a jump cut and a dolly zoom, and how different technical approaches produce different effects. Audiences and critics follow along, recognizing homages and allusions, and appreciating variations and innovations.

Tabletop game design is at a somewhat earlier stage, and this book is our attempt to begin to build a broader game design vocabulary and body of knowledge. Rather than attempting to formalize a specific game design language—an approach that is both daunting and perhaps premature—we've chosen to look at the building blocks of games themselves: the mechanisms or, if you prefer, the mechanics, as they're also frequently called by gamers. (We'll use the terms interchangeably through this book.)

The second-best piece of advice any new designer gets is to play more games. (The first piece of advice, by the way, is to create a physical prototype as quickly as possible.) Playing more games helps designers learn and grow by seeing, first-hand, a large variety of possible game experiences and mechanics. This is no different than any other creative fields: every artist is enriched by experiencing the art and craft of other creators. Yet games, like books, can take a long time to experience, to master, and to fully appreciate, particularly for games that require large player counts, have long play times, or demand many repeated plays.

In an effort to accelerate the learning process, we present this book, which is a compendium of game mechanisms, grouped together thematically, that map the territory of modern gaming. Our goal is not to give a list of steps or instructions for how to design a game. To use a cooking metaphor, this is not a recipe book, but rather a catalog of ingredients, and how they can enrich a dish. We define close to two hundred different mechanisms and variants, spanning topics like Movement, Game End and Victory, Economics, and more. Within each topic, we discuss how different mechanisms create different player experiences, what types of games these mechanisms give rise to, common pitfalls in implementing them, and even some of the physical user-interface issues raised by these mechanisms. With each mechanism, we present illustrative examples and games for further study, running the gamut from modern classics to contemporary new releases, as well as some lesser-known titles that nonetheless have much to teach. A good resource for learning more about the included examples is the website, BoardGameGeek.com, which contains a massive game database featuring images and rulebooks that can assist in further inquiry.

Our hope is that this book serves many purposes. New designers can certainly benefit from reading this book and taking in many mechanisms in a short period of time, but this book can also be used as a reference. Whether you're an experienced designer looking for an overview of auctions, for example, or a gamer interested in exploring the worker placement genre, this book offers an easy way to review the topic and learn more. We feel this book will be of use to educators, students, professional and amateur designers, and anyone interested in reading about game design.

As important as what this book is, is what it is not: a comprehensive listing of all game mechanisms. Though we strove to be broad and inclusive, an exhaustive compilation of all game mechanisms was never our intention, and is arguably impossible. While we do brush lightly on topics like narrative, dexterity, and pantomime, for example, there remains a lot of unmapped terrain. Similarly, while we do sometimes bring in examples from war gaming, miniatures gaming, classic card games and collectible card games, we largely center ourselves on modern board and card games. These lines are certainly artificial, and we have not sought to put forward a definition of what games are in drawing the lines as we did. Rather, we are eager to look to a future that continues to stretch, experiment with and reconsider what games can be in light of new innovations in gaming.

We also do not mean to imply in any way that the well of new mechanisms has run dry. The opportunity for innovation in gaming is limitless.

Our intention is to make it easier for designers to learn to design, to talk with one another about design, and to mine the existing canon of board games for insight into how to design better games. We hope this collection is useful and inspirational to designers, and we look forward to playing the next generation of their games.

A Note on References in This Book

For your convenience, whenever we refer to another part of this book, we include a chapter reference, presented as an abbreviation. For example, the abbreviation ACT is for the chapter about Actions. A reference like ACT-02 means the second section of the Action chapter. This additional information may be helpful to you in understanding the context of the reference, even if you don't choose to look it up right away, and may help in searching for references in digital versions of this text.

1

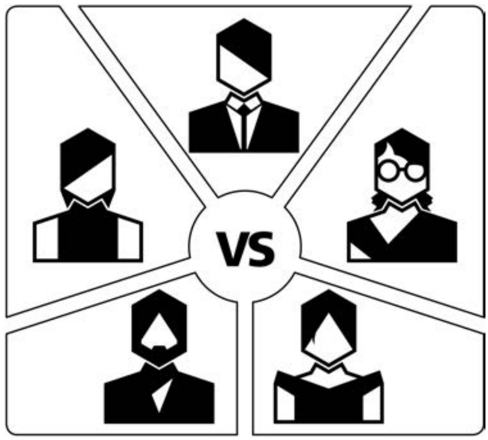
Game Structure

The first choice designers make about a game is the game's basic structure. Who wins? Who loses? What's the overall scope of the game experience? Will it be just one game, or perhaps a series of hands? Maybe the game will encompass many scenarios, with game-state information persisting from play to play? The last 20 years of tabletop gaming have brought us enormous innovation in game structures, with no signs of slowing down.

Game structure raises the question of what it means to actually play the game. In a cooperative game, is the discussion among the players over which action to take the core of the gameplay, or is it the active player executing the action that is core? In games which limit communication, is the main gameplay overcoming the communication limits, or solving the puzzle the game presents? Designers need to be sensitive to participation issues in team games as well, because the game itself may force most of the action over to one teammate at the expense of another.

In this chapter we'll consider both the traditional structures like competitive and team games, as well as the latest ideas in scenario-based, legacy, and consumable games.

STR-01 Competitive Games



Description

A game with two or more players and a single winner.

Discussion

This is the most familiar game structure, the one we encounter as children in games like *Candyland* and *Snakes & Ladders*. Competitive games still make up the large majority of the market for tabletop games. They typically offer a symmetry of expectations: putting aside the impacts of unusual luck and skill, each player begins the game with a roughly equivalent chance at victory. When this promise is broken, the game is considered imbalanced, and may even be tagged with the label "broken." As we'll discuss later (see "Variable Player Powers" in this chapter and "Variable Setup" in Chapter 6), players may have perfectly symmetric factions and starting conditions, or highly asymmetric ones. But in both cases, it's important that the game offers roughly even chances of victory to each player.

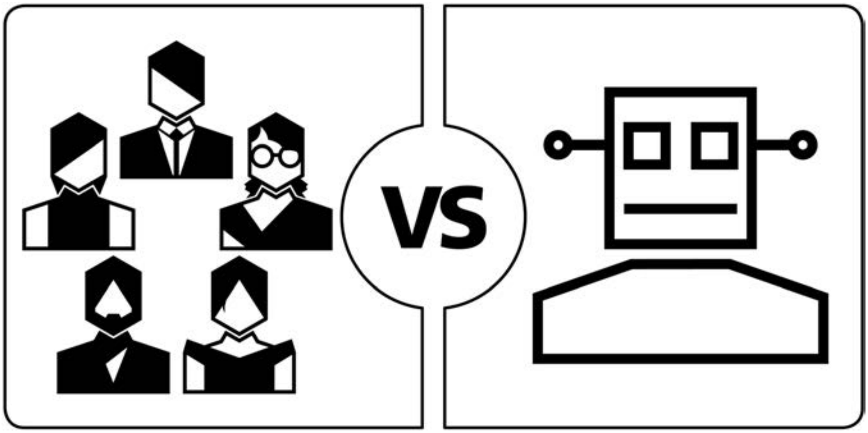
In many games there is an asymmetry which in-game balancing alone can't solve, like the first-mover advantage in chess or the service advantage in tennis. Competitive games often balance these advantages through meta-structures, like tournaments, that offer each player an equal number of chances to play from the advantaged position. In multiplayer games, this may be less practical. Instead, players may be offered opportunities like the bidding in bridge, betting in poker, or the early alliances in *Diplomacy,* to balance any perceived or actual inequities.

The promise of balance in a competitive game gives rise to a number of related issues, like methods for determining victory and breaking ties. Tiebreaking, in particular, is interesting because determining a winner in a competitive game depends upon the game storing information that allows us to decide who played the better game. For many games, this information may not be available. Once we have to look beyond who scored the most victory points, or who crossed some finish line first, there may not be relevant game-state information that could allow us to reasonably determine which of the tied players played best. And yet, in competitive games, many players disdain a game that ends in shared victory. Experience-oriented designers should consider that players tend to recall how a game ended more than the rest of the play experience, and designers should seek to avoid an indecisive conclusion.

Sample Games

Acquire (Sackson, 1964)
Candyland (Abbot, 1949)
Chess (Unknown, ~1200)
Diplomacy (Calhamer, 1959)
Snakes & Ladders (Unknown, ~200 BCE)

STR-02 Cooperative Games



Description

Players coordinate their actions to achieve a common win condition or conditions. Players all win or lose the game together.

Discussion

Quite a few games call for cooperative play among players, including team games, one-vs.-many games, role-playing games, and games with secret traitors. These can be viewed as belonging to a hierarchical category of cooperative games. Some might even include solo games in this group. For our purposes, we'll treat each of these as separate categories, and limit ourselves here to "pure" cooperative games in which all players play on one side and win or lose as a group.

Since 2008, when Matt Leacock released *Pandemic*, the genre of cooperative tabletop games has exploded. Earlier games like *Sherlock Holmes: Consulting Detective, Arkham Horror* and *Lord of the Rings* laid a foundation, and enjoy enduring popularity, but Leacock started a wave of innovation in cooperative gaming that continues to reshape modern gaming a decade and more later.

Cooperative gaming is notable because it lowers barriers to entry for a game. Disparities in skill level can often make a competitive game a sour experience both for the expert and the newcomer. Complex competitive games can be intimidating to new players, and being coached by your opponent introduces some negative play dynamics because of the misaligned incentives of helping your opponent. The power imbalance between the players can also create awkward social dynamics. Cooperative games put players on the same team and help create comradery while allowing experienced players to help teach both the mechanics and strategy of the game without facing conflicting incentives. For many new players, cooperative games are not only a gateway into gaming, but a mainstay of their ongoing consumption of games.

Cooperative games can broadly be placed into two categories: those with an artificial intelligence (AI) and those without. Cooperative games with an AI, like *Sentinels of the Multiverse* and *Mice and Mystics,* feature an opponent or opponents that players must contend against, whose behavior is governed by some kind of simple artificial intelligence, encoded by the designer. In *Sentinels,* the AI is driven by a deck of cards that governs the actions of the enemy villains and the players they will target. *Mice* has a simple algorithm that players use to control the play of enemy figures.

Non-AI games like *Hanabi*, the revolutionary Antoine Bauza title, and *Mysterium,* present players with a puzzle to solve, and some kind of limitations of time, resources, and interaction that players must contend with. However, these games have no villain or opposing force that drives the action and actively confronts the players.

Another distinction between different kinds of cooperative games is whether each player retains agency over their in-game resources, actions and choices, or whether they seek consensus for all decisions, even if they nominally represent separate in-game characters. We might call the former game a partnership game, and the latter a collaborative one. In general, cooperative games will tend to be played collaboratively unless the rules specifically and substantially impede this collaboration and force players to make independent decisions rather than build consensus. Examples include limits on communications, time, and focus.

For some players, collaborative play contributes to the "alpha player problem," also known as "quarterbacking," in which some player takes control of the group discussion and decision-making and creates a negative play experience by overriding other players. There are many possible reasons for the rise of an alpha player problem, many ways that problem can manifest, and a thicket of contributory social dynamics that are beyond the scope of this work. While some players and designers believe that the alpha player problem is a group-composition problem, or a problem of unshared assumptions, rather than a design problem, some design choices will make a game more vulnerable to alpha player takeovers. In particular, when all players share the same information, and the game state is not too complex, the tendency is for alpha player behavior to arise.

At the other end of the spectrum are games which cannot be taken over by an alpha player. *Magic Maze* and others of its type make player communication a game mechanism, such that players can't freely share information or advise one another on how to play. *The Mind* takes this to an extreme by forbidding players from having any kind of communication about which cards they hold. These types of communication limitations may be presented like any other rule, but they don't actually create a bright line of which conduct is and is not permitted. Rather, these games can be played somewhat differently by each group, with the precise contours of allowable communication varying by tacit or overt agreement. This approach is deeply polarizing, and some players will utterly reject these kinds of games or cast doubt on whether they are games at all. That said, these communication restrictions have the potential to create incredible experiences that connect participants to one another on an almost mystical level.

Communication limitations are only one approach to preventing players from achieving consensus-based play. *Space Cadets, Space Alert,* and *FUSE* introduce a real-time element that forces players to make independent decisions because there is no time for players to collaborate. Other games attempt to strongly connect players to their roles, provide them with hidden information, or make operating their roles especially complicated. *Mechs vs. Minions* and *Spirit Island* both make it challenging for players to decipher each other's powers and possibilities. *Sentinels of the Multiverse* attempts something similar by providing each player with a unique preconstructed deck. Escape room games from *T.I.M.E Stories* to the *Exit* and *Unlock* series sometimes bar players from sharing information too specifically as well. The variety of challenges and puzzles these games offer, and even the various roles that players can take in the solving effort, all help ensure that every player finds a satisfying way to participate in the game.

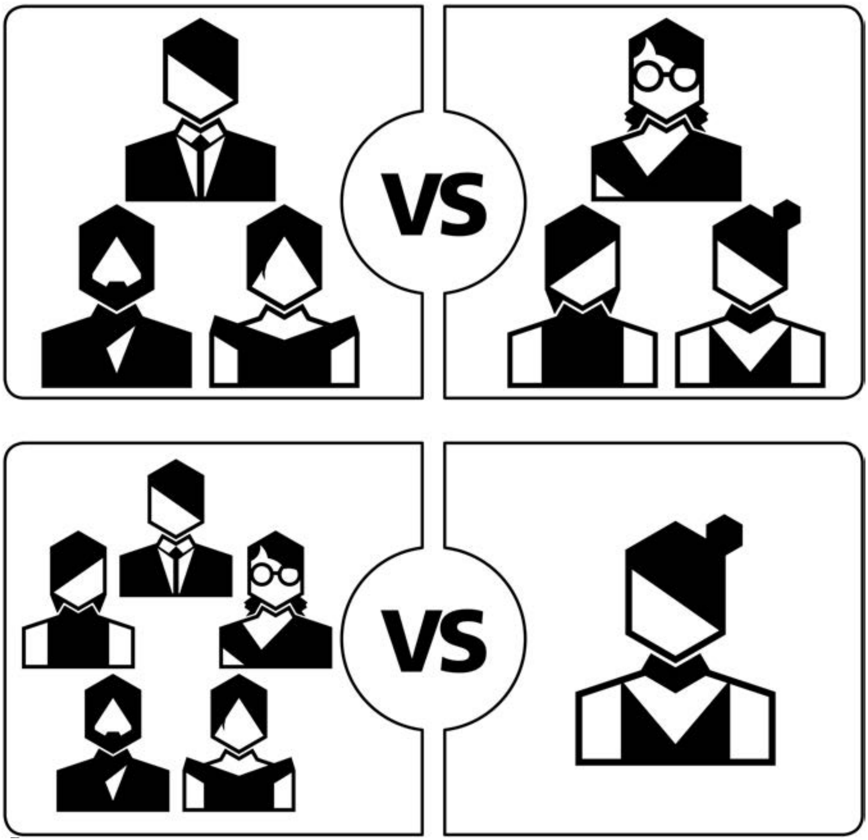
Another notable trend in co-op game design is the conversion of one-vs.-many, "overlord"-style games into co-ops with the assistance of an app. *Mansions of Madness: Second Edition* and *Star Wars: Imperial Assault* both introduced apps that allow the games to be played cooperatively. More generally, games are being released with cooperative and solo modes alongside competitive modes of play. Sometimes, as has been the case with *Orleans* and *Oh My Goods!,* cooperative modes have been introduced in expansions to competitive games.

The ongoing design exploration of cooperative games and their possibilities is one of the most exciting and fruitful trends in tabletop gaming today. Designers are encouraged to experiment with this popular and adaptable game structure.

Sample Games

Arkham Horror (Krank, Launius, Petersen, and Willis, 1987)
Exit: The Game (Brand and Brand, 2016)
FUSE (Klenko, 2015)
Hanabi (Bauza, 2010)
Lord of the Rings (Knizia, 2000)
Magic Maze (Lapp, 2017)
Mansions of Madness (Koneisczka, 2011)
Mechs vs. Minions (Cantrell, Ernst, Librande, Saraswat, and Tiras, 2016)
Mice and Mystics (Hawthorne, 2012)
The Mind (Warsch, 2018)
Mysterium (Nevskiy and Sidorenko, 2015)
Oh My Goods! (Pfister, 2015)
Orleans (Stockhausen, 2014)
Pandemic (Leacock, 2008), and the complete line of *Pandemic* games
Sentinels of the Multiverse (Badell, Bender, and Rebottaro, 2011)
Sherlock Holmes Consulting Detective: The Thames Murders & Other Cases (Edwards, Goldberg, and Grady, 1981)
Space Alert (Chvátil, 2008)
Space Cadets (Engelstein, Engelstein, and Engelstein, 2012)
Spirit Island (Reuss, 2017)
Star Wars: Imperial Assault (Kemppainen, Konieczka, and Ying, 2014)
T.I.M.E Stories (Chassenet and Rozoy, 2015)
Unlock! series (Various, 2017)

STR-03 Team-Based Games



Description

In team-based games, teams of players compete with one another to obtain victory. There are a variety of possible team structures, including symmetrical teams like 2v2 and 3v3, multiple sides like 2v2v2, and even One vs. All.

Discussion

Team-based play is an ancient human pastime and is prevalent in sports, classic card games, and war games. Thematic board games that model some conflict often have to deal with a bilateral narrative, whether in the conflict of good versus evil like in *War of the Ring*, the eponymous conflict of *Axis & Allies*, or the entirely fabricated battle between villagers and their lycanthropic tormentors in *Werewolf.* Team games allow designers to faithfully recreate these two-sided conflicts while making space for more players at the table.

Most team games assign players to one team or another at the outset and allow collaboration and communication among partners. In some cases, partners may share territory and resources, or even move each other's pieces around the board. Early editions of *Axis & Allies,* for example, featured Commander-in-Chief rules to model the historical reality of coordinating attacks by multinational armies.

Assigning players to teams in secret is a common trope in social deduction and betrayal-style games. In *Werewolf,* the werewolves know one another, but the villagers are left to deduce who is a fellow human and who might devour them in the night. In *Battlestar Galactica*, neither humans nor Cylons are revealed to one another, leading to intense suspicion and paranoia. *Battlestar Galactica* features a further twist: team assignments are not static, and a player might start the game as a human, only to become a Cylon in a mid-game loyalty phase that can potentially reassign players from the human to the Cylon side.

Some games support partnering at the meta-game level. In *Risk, Diplomacy,* and many other games, the rules specify that alliances can be formed and broken freely. There are even games that further encode these alliances into the rules. *Dune* provides for different end-game conditions for alliances with different numbers of factions, includes a specific phase when alliances may be formed, and has rules governing the conduct of allies (e.g., they may not attack each other). *Eclipse* establishes economic advantages for allying with other players, while limiting the number of these allegiances, and enforcing in-game penalties on betrayers. There is a great deal of fluidity between games that allow for coordination in play and those that have explicit mechanisms to support that coordination.

Another common approach to team-based games is that in which players perform different roles in the game. This goes well beyond the distinct player abilities common to many co-op games. Party games from *Celebrity* to *Taboo* to *Codenames* feature one player in a clue-giving role, with one or more teammates in a guessing role. This structure is especially suitable for party games because of the concurrent play of the guessers, and the ability for the game structure to support large numbers of players at any one time. It also allows players to easily enter and exit from a game without negatively impacting the overall play experience for other players.

It's not just party games that embrace role separation, though. *Space Cadets: Dice Duel* and *Captain Sonar* both use the setting of operating some kind of ship to cast players as engineers, radar operators, captains, and weapons specialists. Each player plays a mini-game to model their function within the ship, and the broader game is the sum of these coordinated parts. Sometimes these roles can be fixed for the whole game, sometimes players will rotate through the roles every round, and sometimes the game itself will force players to change roles. This last approach can be thematically interesting and appealing to experienced players, but can be especially challenging for new players, who can easily be disoriented when forced to change roles. It also makes teaching these games much more difficult, since, in theory, every player needs to know every mini-game before play can begin. Designers need to take this into account, perhaps by reserving role-changing cards and other triggers for advanced play only.

Somewhat less common are team games with more than two teams competing at once. Skirmish games like *Star Wars: X-Wing* can allow for these types of modes, but there are even more unusual examples. *Cyclades* with the *Titans* expansion enables a 2v2v2 mode, and *Ticket to Ride: Asia* also supports 2v2v2 play.

Team modes are fairly common in otherwise non-team games, and some team modes are attempts to enable a 1v1 game to scale to a higher player count. The aforementioned *Axis & Allies* and *War of the Ring* have team modes, but players may simply choose to play using the 1v1 rules, and have teammates collaborate in decision-making instead of using the formal rules for team play.

Two other game genres feature partnerships. Card games, especially those in the trick-taking family, are one. The other partnership genre is dexterity games. *Flick 'em Up* has explicit partnership rules, and many flicking games like *PitchCar* and *Caveman Curling* are amenable to alternating play.

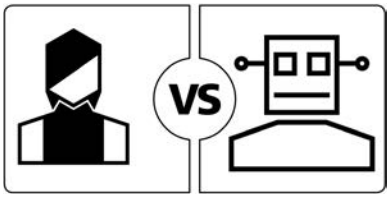
Another very popular game structure is One vs. All. This structure is especially suitable for hidden-movement hunting games, in which one player attempts to escape from a group of hunters, like *Scotland Yard, Specter Ops,* and *Hunt for the Ring.* It's also common in overlord-style games where one player controls the "bad guys" and the other players each control an individual hero. *Conan, Star Wars: Imperial Assault,* and *Level 7 [Omega Protocol]* all feature this style of play. Fantasy Flight Games has converted many of its overlord games into cooperative games by introducing an app to control the overlord player.

What distinguishes One vs. All most sharply from other team games is the asymmetric nature of the factions. In hunting games, each player typically controls only one character, but the characters tend to have very distinct abilities, and the victory conditions for the two sides are different too. In overlord games, the overlord usually controls a whole host of minions that they can deploy, whereas the other players control only one character. These games can be difficult to teach and learn because players need to learn how both factions work in order to play the game effectively. While the genre remains very popular, that challenge continues to bedevil designers and players.

Sample Games

Axis & Allies (Harris, Jr., 1981)
Battlestar Galactica: The Board Game (Konieczka, 2008)
Captain Sonar (Fraga and Lemonier, 2016)
Caveman Curling (Quodbach, 2010)
Celebrity (Unknown)
Codenames (Chvátil, 2015)
Conan (Henry, Bauza, Bernard, Cathala, Croc, Maublanc, and Pouchain, 2016)
Descent: Journeys in the Dark (Second Edition) (Clark, Konieczka, Sadler, and Wilson, 2012)
Diplomacy (Calhamer, 1959)
Dune (Eberle, Kittredge, and Olatka, 1979)
Eclipse (Tahkokallio, 2011)
Flick 'em Up (Beaujannot and Monpertuis, 2015)
Hunt for the Ring (Maggi, Mari, and Nepitello, 2017)
Level 7 [Omega Protocol] (Schoonover, 2013)
Mansions of Madness: Second Edition (Valens, 2016)
PitchCar (du Poël, 1995)
Risk (Lamorisse and Levin, 1959)
Scotland Yard (Burggraf, Garrels, Hoermann, Ifland, Scheerer, and Schlegel, 1983)
Space Cadets: Dice Duel (Engelstein and Engelstein, 2013)
Specter Ops (Matsuuchi, 2015)
Star Wars: Imperial Assault (Kemppainen, Konieczka, and Ying, 2014)
Taboo (Hersch, 1989)
War of the Ring (Second Edition) (Di Meglio, Maggi, and Nepitello, 2012)

STR-04 Solo Games



Description

A solo game is a game or game mode intended for play by a single player.

Discussion

Solo games can stretch the definition of what a game is, but solo games have been around for some time. Patience or Solitaire card games, war games, interactive fiction (e.g., Choose Your Own Adventure® books), crossword puzzles, paper-and-pencil games, word searches and many more single-player, game-like pursuits are very popular and have been for years. What is new is the explosion of demand, and supply, for solo board games, whether as standalone games or as game modes in multiplayer games. Rather than quibbling over the definition, we choose to embrace the variety and possibilities of solo gaming.

Solo games are closely related to cooperative games in the sense that the player plays against the game system itself, rather than a sentient opponent. Solo games can be considered as a special case of co-op games in which there is only one player—and thus, no alpha player problem. While most co-ops can be, and are, played solo, communications-based co-op games like *The Mind* typically can't be played solo at all. Cooperative Limited Card Games (LCGs—a genre, i.e., a blend of collectible card games and deck-building games) are very popular among solo players, and the dungeon-crawl-style Rogue-like game seems especially amenable to solo play. Players can usually brew their own unofficial solo game from these games, or from many non-communication-based co-ops, by playing more than one position in a co-op game, for example.

Modern solo design pushes beyond that basic approach to solo play. We can classify a few kinds of solo games: goal-based, record-based, and AI-based. In goal-based games, players try to achieve some goal, usually a Victory Point (VP) total, within a certain number of turns, or a certain amount of time.

This type of design is most similar to multiplayer cooperative games. These games might have an AI opponent of sorts, but typically the opponent is an asymmetric villain of some kind, or even an entirely abstract process, like a spreading fire in *Flash Point: Fire Rescue*. The AI is not an automated version of a human player who competes as another human player might.

Record-based games are those whose goal is to beat your previous high score. While this may seem like a mundane and dull victory condition, some games have developed large solitaire followings, complete with ladders and statistics that transform the game from a solo game to a community-based meta-competition. *Ganz schön clever* is one example of this type of game and community competition. Some goal-based games can also work as record-based games, as long as some scoring metric can be captured that adequately measures game performance.

AI-based games attempt to recreate the multiplayer experience by introducing an automated player or players. Typically, this means some basic algorithm governs the moves of the AI player. These systems can be challenging to design, and usually depend on dice, cards or other randomizers to create uncertainty about the AI player's choices. However, other approaches to AI opponents are also possible, as shown by Morten Monrad Pederson's Automa system, which powers solo modes in games as varied as *Scythe, Anachrony, Baseball Highlights 2045,* and *Between Two Cities.*

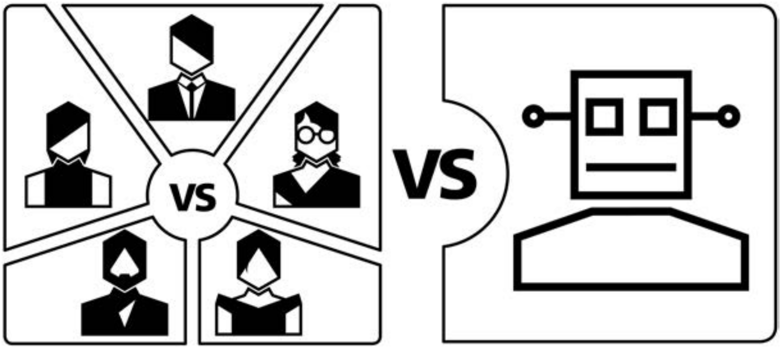
Pederson's Automa system is more a philosophy and approach than an official system, and many designers have used its principles to create Automas. The key principles are that the solo game should feel as much as possible like the multiplayer game, and that AIs should be representational, rather than procedural. The AI doesn't observe the rules and doesn't play the game as a player might. Rather, the AI is a series of outputs that are applied to the game state that mimic the outputs that result from a human player's choices—a kind of Potemkin player. In Pederson's words, the AI is only the shell of a human player, and while the humans in a game of *Viticulture* have vineyards and grow grapes, the Automa players do not; they just give the impression that they do. In practice, Automas will claim action spaces, draft cards, block routes, and otherwise interfere with human players, but they will not collect resources, build buildings, or score victory points.

Solo games are enjoyed by a large, active, and growing community who continues to homebrew solo versions of multiplayer games, host contests and challenges, and enjoy an unexpected comradery despite the solitary nature of their pastime. Increasingly, including a solo mode is a desirable feature for published multiplayer games, and the success of a number of solo-only games, like *Friday* and *Hostage Negotiator,* has established that there is a market even for the loneliest of player counts.

Sample Games

Ambush (Butterfield and Smith, 1983)
Anachrony (Amann, Peter, and Turczi, 2017)
Arkham Horror: The Card Game (French and Newman, 2016)
B-17: Queen of the Skies (Frank and Shelley, 1981)
Baseball Highlights: 2045 (Fitzgerald, 2015)
Between Two Cities (O'Malley, Pedersen, and Rosset, 2015)
Flash Point: Fire Rescue (Lanzig, 2011)
Friday (Friese, 2011)
Ganz schön clever (Warsch, 2018)
Hostage Negotiator (Porfirio, 2015)
Lord of the Rings: The Card Game (French, 2011)
Mage Knight Board Game (Chvátil, 2011)
One Deck Dungeon (Cieslik, 2016)
Onirim (Torbey, 2010)
Robinson Crusoe: Adventures on the Cursed Island (Trzewiczek, 2012)
Scythe (Stegmaier, 2016)
Terraforming Mars (Fryxelius, 2016)

STR-05 Semi-Cooperative Games



Description

A game which ends with either no winners or with the players winning as a group, but a single player being recognized as the individual winner as well.

Discussion

Semi-cooperative games are like cooperative games because players can all lose together to the game itself. However, if the players manage to overcome the game, one of the players will be crowned the individual victor, or a kind of most valuable player, based on some in-game achievement. A group's enjoyment of this style of game depends on all members having the same answer to the question of whether a group win, coupled with an individual loss, is in fact superior to a total group loss. If players are split on this question, those for whom an individual loss is as bad as a group loss will not seek optimal cooperative plays, and may even sabotage the group while trying to secure a winning individual position. Players who prefer to win as a group, even if they lose as individuals, will be incensed by this conduct, and unhappiness will follow for everyone.

There's a strong connection between semi-cooperative games and role-playing games (RPGs). RPGs typically feature a similar dynamic: all players want to follow the main plot narrative and defeat the big bad, but each player has individual motivations that may sometimes work at cross-purposes, and in any case, each player is motivated to secure the greatest in-game rewards in terms of wealth, experience points, glory, etc. These competing motivations can help weave a complex and gripping narrative. In tabletop games, a semi-cooperative structure can create memorable moments of last-second betrayal and competitive jockeying for position.

Semi-cooperative games are fascinating as a design study because of how players engage with them. Many groups will play semi-cooperatives as co-ops, ignoring the individual win conditions, or treating them as a means for recognizing a great performance by some player, rather than as a win condition to be claimed. Others will embrace the notion of an individual win and engage in brinksmanship in almost total violation of the cooperative element of the game. Despite the broad range of possible play styles, each group typically has a very narrow idea of the "right way" to play the game. In other words, it's unlikely that the same group would play *Arkham Horror* as a total co-op one night, and then as a semi-co-op another night. Actually, you might be hard-pressed to find anyone who plays *Arkham Horror* with any care for the individual win condition. The game holds little attraction for players looking for that experience. Crafting a semi-co-op that is satisfying to play and that manages to incentivize both the individual and cooperative aspects of the game is like trying to land a jet plane in a phone booth.

Another route a designer can consider is introducing cooperative elements into a competitive game without creating a cooperative loss condition. Games like *Kingsburg* and *Survive: Escape from Atlantis* achieve this in different ways. In *Kingsburg,* all players must face a common foe at the end of each year. In *Survive,* players can share common resources like lifeboats to escape the ocean's dangers. These cooperative opportunities can enrich the game and create internal drama without confusing players as to their incentives, or causing distress over what type of game players they thought they sat down to.

Sample Games

Archipelago (Boelinger, 2012)
Arkham Horror (Krank, Launius, Petersen, and Willis, 1987)
Castle Panic (De Witt, 2009)
CO_2 (Lacerda, 2012)
Defenders of the Realm (Launius, 2010)
Kingsburg (Chiarvesio and Iennaco, 2007)
Legendary Encounters family of deck-building games (Cichoski and Mandel, 2014)
The Omega Virus (Gray, 1992)
Republic of Rome (Berthold, Greenwood, and Haines, 1990)
Survive: Escape From Atlantis (Courtland-Smith, 1982)

STR-06 Single Loser Games

Description

A game with three or more players which ends with only one loser.

Discussion

Games with one loser flip the script by incentivizing players not to perform better than everyone else, but better than at least one other person. Like the old saw about not needing to outrun a ravenous bear, these games emphasize impeding other players more than advancing yourself. *Old Maid* is a well-known classic card with this structure. *Alcatraz: The Scapegoat* is a modern game in which players are escaping prison, but one of them, the scapegoat, will remain behind and lose.

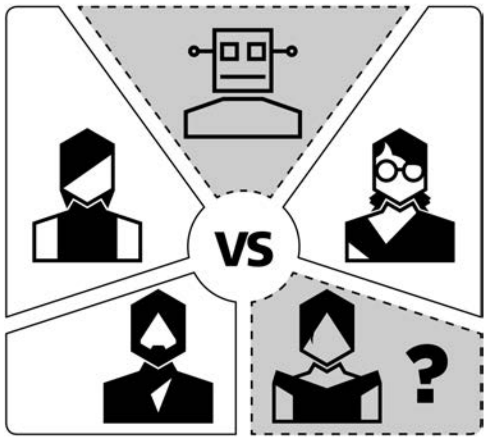
Stacking games often feature a single-loser structure, because many of them end when a player knocks over the shared structure. *Jenga* and *Rhino Hero* are examples of this dynamic, though *Rhino Hero* does offer some rules for determining who the winner is among the surviving players. In practice though, groups often ignore this final scoring, or even scoring of any kind.

Single-loser structures are not very common in modern games, perhaps because they encourage sharply confrontational "take-that" play in which players can directly impede or harm one another. These structures can also be susceptible to a bash-the-loser pattern, in which the player who falls behind becomes the most attractive target to all other players, and has no chance to recover their position. This structure represents an innovation horizon for designers to explore.

Sample Games

Alcatraz: The Scapegoat (Cywicki, Cywicki, and Hanusz, 2011)
Aye, Dark Overlord (Bonifacio, Crosa, Enrico, Ferlito, and Uren, 2005)
Cockroach Poker (Zeimet, 2004)
Jenga (Scott, 1983)
Pairs (Ernest, Glumpler, and Peterson, 2014)
Rhino Hero (Frisco and Strumph, 2011)

STR-07 Traitor Games



Description

A traitor game can be seen as a kind of team game, or as a cooperative game with a betrayal mechanism. The traitors typically win by triggering a failure condition for the players. For our purposes, a traitor game is characterized by traitors that begin the game with hidden identities.

Discussion

Traitor games sit at the overlap in the Venn diagram of social deduction games, team games, and cooperative games. The cooperative game engine sets the overall parameters for how players interact in the game, and those choices provide information that players can use and manipulate to deduce, accuse, and mislead. The team element provides coordination options for both sides, both out loud and in secret. These designs must add a layer of obfuscation in order to preserve uncertainty and mystery. Thus, in *Dead of Winter* the cards contributed by players to meeting the current crisis are shuffled prior to being revealed, so as to hide which player contributed which card.

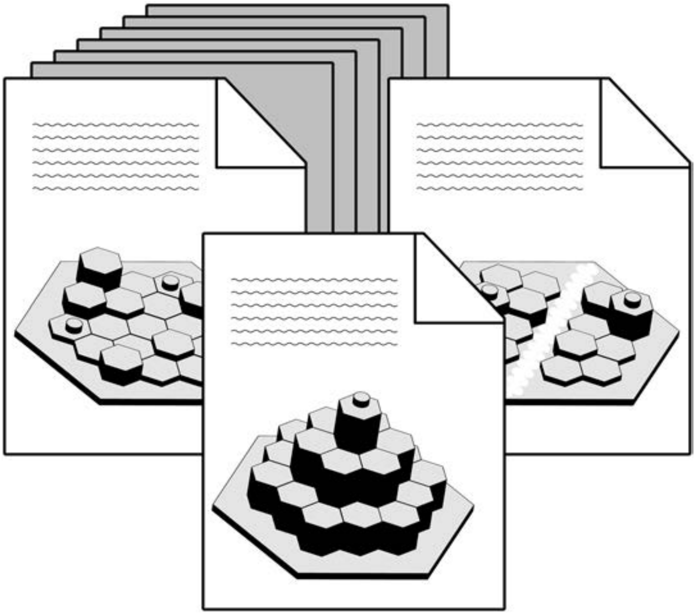
In designing traitor games, it's important to consider how the game plays at different stages of knowing the traitor's identity. The tension and suspense of not knowing who is on which side is great fuel for the beginning of the game, but once the traitor has been sussed out, is the game still fun? Does the traitor still have meaningful and interesting choices? Is catching the traitor a victory condition itself? *Betrayal at House on the Hill* is a traitor game with many story modules, each with its own mechanisms, that designers can treat it as a case study on this exact issue.

Another challenge facing traitor games is learnability. Like in team or One vs. All games, players have to learn how all the sides work to play properly. In traitor games with hidden identities, players can't readily ask questions or consult the rules without giving away their loyalties. Good player aids and reference material need to be areas of focus.

Sample Games

Battlestar Galactica: The Board Game (Konieczka, 2008)
Betrayal at House on the Hill (Daviau, Glassco, McQuilian, Selinker, and Woodruff, 2004)
Dead of Winter: A Crossroads Game (Gilmour and Vega, 2014)
Saboteur (Moyersoen, 2004)
Shadows over Camelot (Cathala and Laget, 2005)
Werewolf (Davidoff and Plotkin, 1986)

STR-08 Scenario/Mission/Campaign Games



Description

This is a game system that can be applied to a variety of different maps, starting resources and positions, and even different win and loss conditions. These variable conditions can be assembled into a broader narrative or campaign, or they can be entirely disconnected from one another.

Discussion

The roots of scenario-based games are in storytelling, history, wargames and RPGs. The idea is that a rich experience can be woven out of multiple episodes, planned out to be experienced sequentially.

In wargames like *Advanced Squad Leader,* the same core system, together with substantial supplements, enables players to model an enormous variety of possible conflicts. As players gain mastery, they can attempt more challenging and complex scenarios. At the other end of the spectrum, a war-themed game like *Memoir '44* gives players the opportunity to play the many battles and confrontations that made up the D-Day invasion with an intuitive ruleset that can be taught in ten minutes.

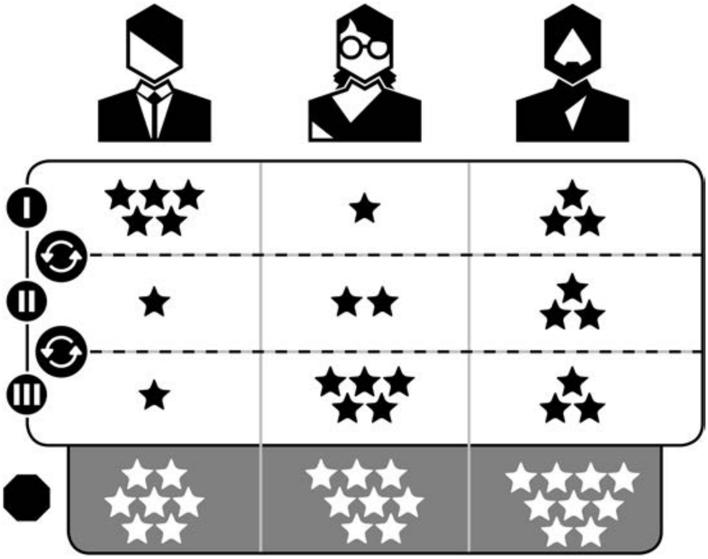
Scenario structures are popular in dungeon crawler games. Swap in a new dungeon map, a new boss monster, maybe some new mission types, and players can enjoy new challenges without learning new rules. Scenarios can also be completely detached from any narrative, and simply allow players to face off against one another in a skirmish mode that provides some basic rules for force construction. Scenarios are also prevalent in cooperative settings, both for offering different levels of difficulty and for creating variety.

Recently, there has been an explosion in the types of games offering a scenario or campaign structure. *Catan* expansions going back as far as *Seafarers of Catan* have offered scenario play, but hits like *The 7th Continent, Mechs vs. Minions, Near and Far,* and card-based dungeon crawlers like *Pathfinder Adventure Card Game* and *Gloomhaven* have rocketed in popularity. It's also impossible not to mention *T.I.M.E Stories* and the various escape room game series like *Exit: The Game, Unlock!,* and *Deckscape.* Though each box is a self-contained experience, the play system is largely similar from box to box, and outside of the packaging and business model, these games are scenario-based in structure. One might even consider *Age of Steam* maps, *Power Grid* expansions, *Ticket to Ride* standalone variants from Europe to New York, *Pandemic: Iberia* and *Cthullhu,* and the *Mystery Rummy* series as follow-on scenarios, sold separately.

Sample Games

The 7th Continent (Roudy and Sautter, 2017)
Advanced Squad Leader (Greenwood, 1985)
Age of Steam (Wallace, 2002)
Dead of Winter: A Crossroads Game (Gilmour and Vega, 2014)
Deckscape (Chiachierra and Sorrentino, 2017)
Exit: The Game (Brand and Brand, 2016)
Mechs vs. Minions (Cantrell, Ernst, Librande, Saraswat, and Tiras, 2016)
Memoir '44 (Borg, 2004)
Near and Far (Laukat, 2017)
Pandemic game system (Leacock, 2008)
Pathfinder Adventure Card Game (Selinker, Brown, O'Connor, Peterson, and Weidling, 2013)
T.I.M.E Stories (Chassenet and Rozoy, 2015)
Ticket To Ride (Moon, 2004)
Unlock! series (Various, 2017)

STR-09 Score-and-Reset Games



Description

This is a game in which players play until reaching a stopping condition, then record scores, reset the game, and play one or more additional rounds. The game concludes after some number of rounds, and the cumulative score is calculated to determine a winner.

Discussion

Score-and-reset is a very common game structure that is well represented among classic games from *Bridge* to *Backgammon,* and in modern games like *Red7, Ravenous River,* and *Incan Gold.*

Card games are especially amenable to this structure, which matches well with playing out a hand of cards, shuffling and redealing. Games with a strong turn-order advantage can take advantage of a score-and-reset structure by having players take turns in the lead position, and using cumulative scoring to measure who played best over a series of rounds.

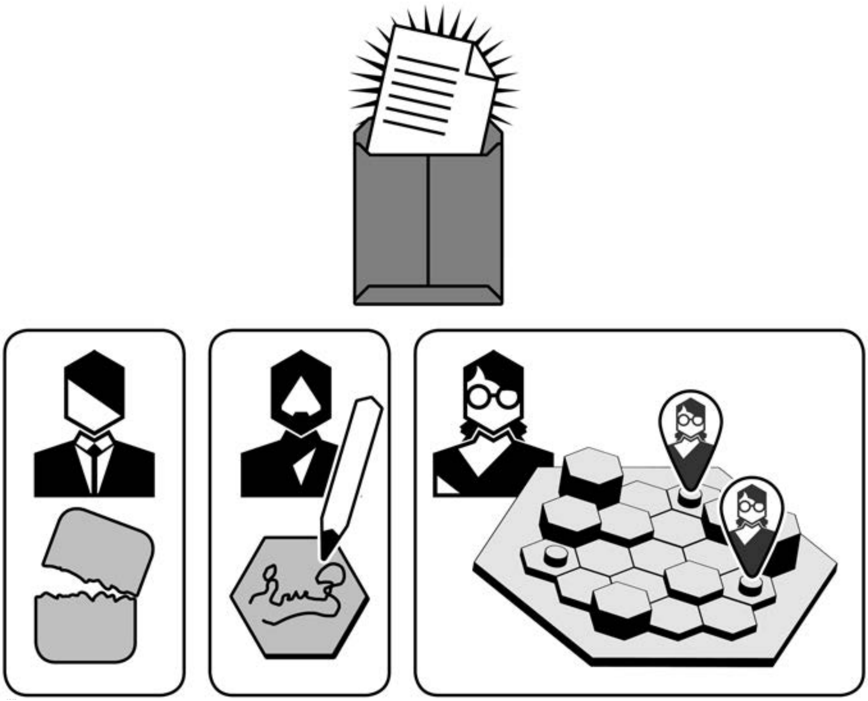
Stacking games like *Rhino Hero* use score-and-reset because once the shared structure is toppled, there's a need to collect up the cards and pieces anyway, and this represents a natural end point. Whether players do in fact record wins and losses or points is to some extent a matter of the dynamics of the game group itself. By contrast, dexterity games like *KLASK* and *BONK* reset after each goal, but scoring is rigorously tracked.

Score-and-reset structures can be somewhat informal. Players might play a series of games of *Hive* and declare a victor after one player reaches some total number of victories. There's not a sharp line between score-and-reset and round structure in general. *Amun-Re* features two ages, in which gameplay is identical, but the board itself does not fully reset. Pyramids built in the first age persist in the second age, and influence valuation of territories. In *Blue Lagoon,* the board is reset between the first and second half of the game, with the exception of the huts players lay out in the first half. However, the rules of placement change profoundly too—where players were free to lay explorers on any sea space in the first half, in the second half, they must begin laying explorers adjacent to their huts.

Sample Games

Amun-Re (Knizia, 2003)
Blue Lagoon (Knizia, 2018)
BONK (Harvey, 2017)
Hive (Yianni, 2001)
Incan Gold (Faidutti and Moon, 2005)
KLASK (Bertelsen, 2014)
Ravenous River (Shalev, 2016)
Red7 (Chudyk and Cieslik, 2014)
Rhino Hero (Frisco and Strumph, 2011)

STR-10 Legacy Games



Description

A legacy game is a multisession game in which permanent and irreversible changes to the game state carry over from session to session.

Discussion

The first legacy game, Rob Daviau's *Risk Legacy*, came from the unlikeliest of places: Hasbro. Daviau, by his telling, half-shepherded, half-snuck the audacious design past the gatekeepers at the company, and emerged with a genre-defining hit. Famously, the game was sealed with a sticker that had to be broken to open the box, and which warned ominously, "What is Done Cannot Be Undone." This irreversible permanent change is what legacy games are all about.

As of this writing, there are still relatively few legacy games on the market. They are difficult to playtest, require generating quite a lot of content, and are tricky to produce and price properly. Curmudgeons will rightly point out that RPGs and campaign games are essentially legacy games, except that you don't have to destroy your game as you play. Yet that permanent destruction, the tearing of a card or writing on a board generates a visceral response that those other games don't. Breaking this taboo and permanently altering a game can be a stressful and cathartic experience at the same time.

Legacy games aren't simply about destroying components and defacing boards. Another element common to these games is unlocks: gated content that can only be opened and accessed after some condition is met. We'll refrain from specific examples, so as not to spoil these experiences for the uninitiated! Unlocks differ fundamentally from gated content in non-legacy games (see "Gating and Unlocking" in Chapter 3) in that unlocks usually occur at game-end, their impacts will only be felt in the next session, and these unlocks are typically not simply buffs and de-buffs to existing statistics, but rather, entire new mechanisms, characters, factions, maps, etc. These unlocks can also radically change the narrative or move the camera and offer players new perspective and surprises.

Like campaign games, legacy games require a substantial commitment to complete, and they typically call for the same group to come together ten or more times. *Pandemic Legacy* can theoretically take 24 sessions to complete! Publishers, and thus designers, can feel trapped between offering a novel and essentially unrepeatable experience on the one hand, while still providing sufficient re-playability to players, despite the consumable nature of the game. Most legacy games do allow players to keep playing the game even after the campaign is concluded, but anecdotally, it seems that few players do so.

It appears, from this early vantage point on this emerging trend, that the market is expanding to allow for a broader range of possible games, such that at the right price point, even a single-play game like the *Exit* series can be a hit. Games like the *Harry Potter: Hogwarts Battle* deck-building game have used unlocks, but not destructibility, and are fully resettable. *Gloomhaven* takes a similar approach with reusable stickers to provide changes that are campaign-permanent, but otherwise reversible. *Fabled Fruit* is even more easily reset, requiring nothing more than reordering a deck of cards. However, it is difficult to distinguish it structurally from, say, a deck-building game in which any given play session may include a completely different subset of cards.

Legacy games are a still-emerging category and definition. Elements we've identified may not exist in every game, and new elements may yet emerge, especially as legacy games and digital apps converge. Stay tuned for more changes in this exciting new genre.

Sample Games

Betrayal Legacy (Daviau, Cohen, Honeycutt, Miller, Neff, and Veen, 2018)
Charterstone (Stegmaier, 2017)
Fabled Fruit (Friese, 2016)
Gloomhaven (Childres, 2017)
Harry Potter: Hogwarts Battle (Forrest-Pruzan Creative, Mandell and Wolf, 2016)
Legacy of Dragonholt (Valens, Clark, Flanders, and Mitsoda, 2017)
Pandemic Legacy: Season 1 (Daviau and Leacock, 2015)
Risk Legacy (Daviau and Dupuis, 2011)
Seafall (Daviau and Honeycutt, 2016)